

SINGLE-PHASE GAS FLOW IN MICROCHANNELS

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2.4. Pulsed gas flows in microchannels

Rarefaction affects steady microflows and modifies the behavior of unsteady microflows. The case of pulsed flows is of particular interest because such flows are encountered in many applications. For example, micropump actuators often generate sinusoidal pressure fluctuations inside a chamber in order to induce a flow in microchannels through microvalves or microdiodes.

Detailed models for pulsed gaseous flows in the slip flow regime can be found in Caen et al. (1996) for circular microtubes or in Colin et al. (1998a) for rectangular microchannels. They are based on the Navier–Stokes equations with first-order slip and temperature jump conditions at the walls. If we consider a microchannel of length l , whose inlet is submitted to a sinusoidal pressure fluctuation with a small amplitude, the gain Δp^* of the microchannel, that is the ratio of the outlet Δp_o over inlet Δp_i fluctuating pressures amplitudes, can be calculated. For example, in the simple case of a microchannel closed at its outlet, this gain takes the form

$$\Delta p^* = \left| \frac{\Delta p_o}{\Delta p_i} \right| = \left| \frac{1}{\cosh(\beta l)} \right| \quad (2.161)$$

where $\beta(f, Kn) \approx \sqrt{f} \psi(Kn)$ depends on the square root of the frequency f and on the Knudsen number. It can be shown that the band pass of the microchannel is underestimated when slip at the walls is not taken into account (Fig. 2.23).

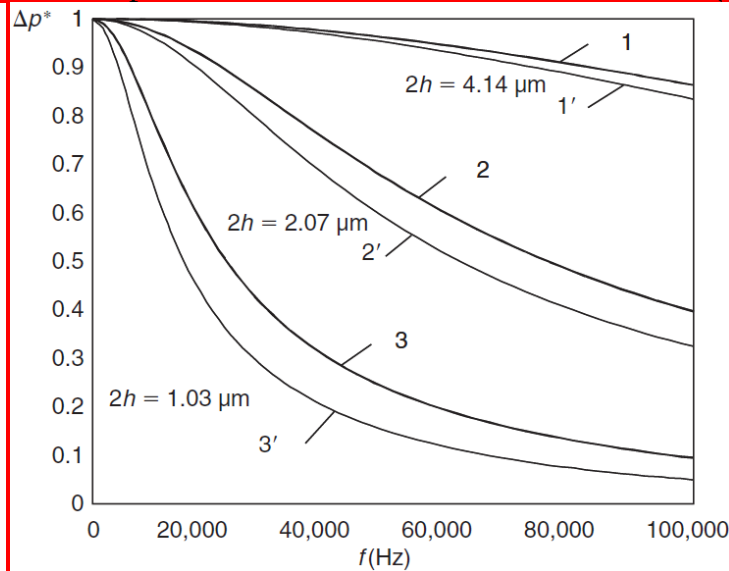


Fig. 2.23. Influence of microchannel depth $2h$ on the gain Δp^* for a square microchannel closed at its outlet. $l = 100 \mu\text{m}$, $\bar{T} = 293 \text{ K}$, $\bar{p} = 0.11 \text{ MPa}$. 1, 2, 3: slip flow; 1', 2', 3' : no-slip flow.

Obtaining experimental data to discuss these theoretical results is very hard, due to the very small size of the pressure sensors required for these experiments. However, experimental data were obtained for microtubes down to $50 \mu\text{m}$ in diameter, using a commercially available pressure microsensor placed in a

minichannel connected in series to one or several parallel microtubes (Colin et al., 1998b). The behavior of the micro- and minichannels association was simulated and compared to these data. The agreement was good, both for the gain and the phase of the association.

The model for pulsed flows in constant-section microchannels can be easily extended to microchannels with slowly varying cross-sections (Aubert et al., 1998). It allows an understanding of the behavior of microdiffusers subjected to sinusoidal pressure fluctuations and to test the diode effect of a microdiffuser/nozzle placed in a microchannel that is subjected to sinusoidal pressure fluctuations. In Fig. 2.24, two layouts (a) and (b) are considered.

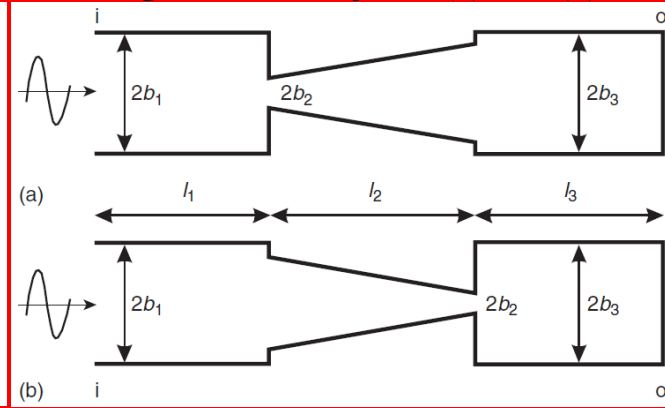


Fig. 2.24. Two layouts of a microdiode placed in a microchannel.

In layout (a), the tested element is a microdiffuser, with an increasing section from inlet to outlet. In layout (b), the same element is used as a nozzle with a decreasing section from inlet to outlet. To obtain an exploitable comparison between the two layouts, the widths ($2b_1$, $2b_2$, $2b_3$) and the lengths (l_1 , l_2 , l_3) are the same in layouts (a) and (b). The angle 2α is also the same for the diffuser or the nozzle.

An analysis of the frequency behavior for each layout shows significant differences, which is characteristic of a dynamic diode effect. Therefore, in order to characterize the dissymmetry of the transmission of the pressure fluctuations, an efficiency E of the diode is introduced. This efficiency

$$E = \frac{\Delta p_{(a)}^*}{\Delta p_{(b)}^*} = \frac{|\Delta p_o / \Delta p_i|_{(a)}}{|\Delta p_o / \Delta p_i|_{(b)}} \quad (2.162)$$

defined as the ratio of the fluctuating pressure gain in layout (a) over the fluctuating pressure gain in layout (b), can be studied as a function of the frequency f . Some results are shown in Fig. 2.25, for different values of the depth $2h$ with $l_1 = l_2 = l_3 = 3$ mm, $2b_1 = 2b_3 = 467$ μ m, $2b_2 = 100$ μ m and $\alpha = 3.5^\circ$.

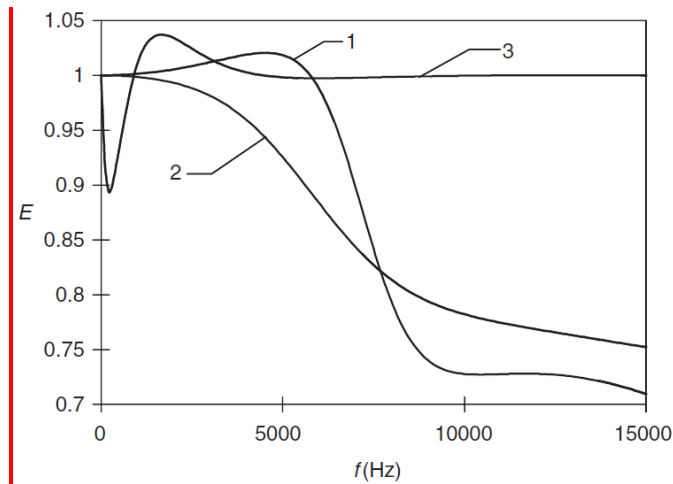


Fig. 2.25. Diode efficiency for different values of the depth. 1: $2h = 100 \mu\text{m}$; 2: $2h = 62 \mu\text{m}$, 3: $2h = 6 \mu\text{m}$.

Figure 2.25 points out that with a microdiffuser ($2h = 6 \mu\text{m}$), E appears to be less than unity below a critical frequency. This denotes a reversed diode effect compared to the case of a diffuser with sub-millimetric dimensions ($2h = 100 \mu\text{m}$), for which E is less than unity beyond a critical frequency. Note also that the influence of slip (taken into account in Fig. 2.25) is not negligible when $2h = 6 \mu\text{m}$. However, slip has little influence on the values of characteristic frequencies (i.e. the frequencies for which E is an extremum or equal to unity).

Pulsed gaseous flows are also of great interest in short microchannels, or micro-orifices, for example in synthetic microjets. The synthetic microjet is a low power, highly compact microfluidic device that has potential for application in boundary layer control (Lee et al., 2003). The oscillation of an actuated membrane at the bottom of a cavity generates a pulsed flow through a micro-slit or orifice (Fig. 2.26). It results in a zero net-mass flux, but in a non-zero mean momentum flux. A series of such microjets can be used to control separation or transition in the boundary layer of the flow on a wing in order to reduce drag, to increase lift or to limit noise generated by the air flow.

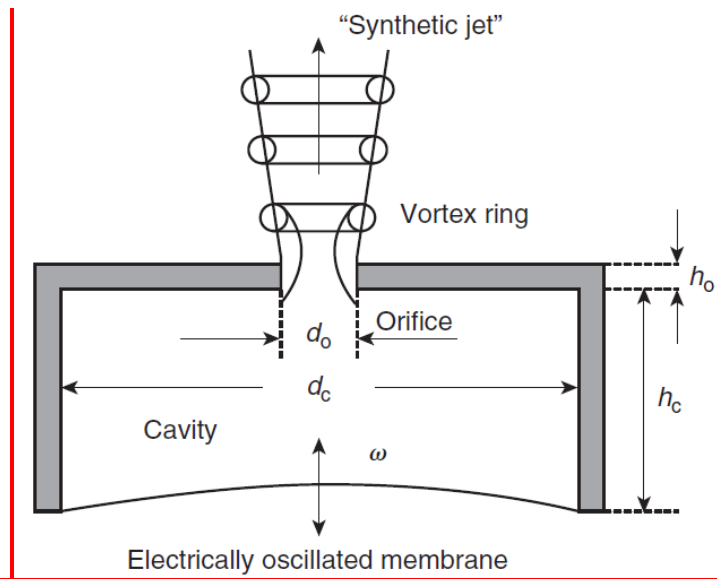


Fig. 2.26. Schematic of a micro-synthetic jet actuator (Mallinson et al., 2003).

2.5. Thermally driven gas microflows and vacuum generation

Generating vacuum by means of microsystems concerns various applications such as the taking of biological or chemical samples, or the control of the vacuum level in the neighborhood of some specific microsystems during working.

The properties of rarefied flows (due to both low pressure and small dimensions) allow unusual pumping techniques.

For high Knudsen numbers, the flow may be generated without any moving mechanical component by only using thermal actuation, which is not possible with classical macropumps.

2.5.1. Transpiration pumping

(Pump: Pressurizes liquids Compressor: Pressurizes gases)

The title is better to be: **Transpiration Compressing** (LK)

Currently, the most studied technique is based on **thermal transpiration**. The basic principle requires **two chambers** filled with **gas** and **linked** with **an orifice** whose hydraulic diameter is small compared with the mean free path of the molecules. **Chamber 1 is heated**, for example with an element, so that $T_1 > T_2$. By analyzing the probability that some molecules move from one chamber across the orifice, it can be shown that if the pressure is uniform ($p_1 = p_2 = p$), **a molecular flux**

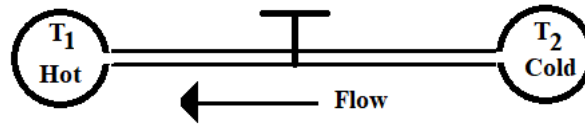


Fig. by LK

$$\dot{N}_{2 \rightarrow 1} = \frac{Ap}{\sqrt{2\pi m k_B}} \frac{\sqrt{T_1} - \sqrt{T_2}}{\sqrt{T_1 T_2}} \quad (2.163)$$

from chamber 2 toward chamber 1 – **from cold to hot temperatures** – appears (Muntz and Vargo, 2002; Lengrand and Elizarova, 2004). If the net molecular flux is constant, the pressures necessarily verify that

$$\frac{p_1}{p_2} = \sqrt{\frac{T_1}{T_2}} \quad (2.164)$$

In Eqs. (2.163) and (2.164), m is the mass of a molecule and A is the area of the orifice. If $p_2 < p_1 < p_2 \sqrt{T_1/T_2}$, there is a net flow from 2 toward 1, which results in a decrease of p_2 and/or an increase of p_1 . **This indicates that a basic microscale pump is working.**

Thus, a Knudsen compressor (Fig. 2.27) can be designed by connecting a series of chambers with very small orifices that have a cold region (temperature T_2) on one side and a hot region (temperature T_1) on the other side by means of an adequate local heater placed just downstream of the orifices. This multistage layout leads to important cumulated pressure drops, while keeping a satisfactory flow rate. In practice, the orifices must be replaced by microchannels and this complicates the modeling (Vargo and Muntz, 1997). Moreover, it is generally difficult to maintain simultaneously a free molecular regime in the microchannels and a continuum regime in the chambers, which are the required conditions for the optimal efficiency corresponding to Eqs. (2.163) and (2.164). More elaborate models, based on the linearization of the Boltzmann equation (Loyalka and Hamoodi, 1990), are able to take into account the transition regime ($0.05 < \text{Kn} < 10$) both in the microchannels and in the chambers (Vargo and Muntz, 1999). Several design studies are proposed in the literature that show the theoretical feasibility of microscale thermal transpiration pumps.

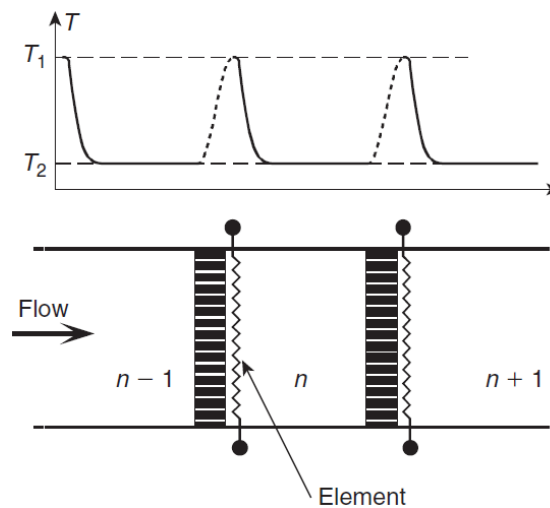


Fig. 2.27. Multistage Knudsen compressor.

The first mesoscale prototypes, heated with an element at the wall (Vargo and Muntz, 1997; Vargo and Muntz, 1999; Vargo et al., 1999), were tested. An alternative solution suggesting heating within the gas was also proposed (Young, 1999), but experimental data are not available for this layout.

Actually, there is yet much to do for the theoretical optimization and almost everything is to be done for the design of thermal transpiration micropumps using micro manufacturing techniques. The performances of these micropumps remain limited for technological reasons. A high vacuum level may become incompatible with the typical internal sizes of microsystems (Muntz and Vargo, 2002) because the regime in the chambers must be close to a continuum regime, which requires sizes too big for the lower pressures.

2.5.2. Accommodation pumping

(Pump: Pressurizes liquids Compressor: Pressurizes gases)

The title is better to be: **Accommodation Compressing** LK)

Accommodation pumping is another pumping technique.

It exploits the property of gas molecules whose reflection on specular walls ($\sigma = 0$) LK) depends on their temperature.

If the wall is warmer than the gas, the mean reflection angle is greater than the incident angle.

Conversely, if the wall is colder than the molecules, they have a more tangential reflection.

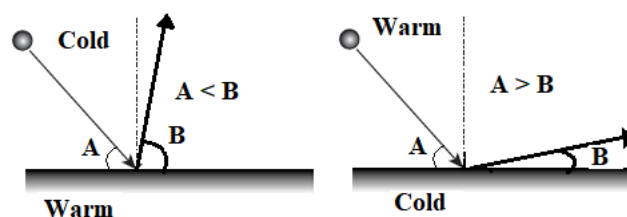


Fig. by LK

Consequently, in a microchannel with perfectly specular walls connecting two chambers at different temperatures, a flow takes place from the hot toward the cold chambers (Hobson, 1970). If the walls of the microchannel give a diffuse reflection, this effect disappears.

This property was confirmed with numerical simulations by the DSMC method (Hudson and Bartel, 1999). By linking two warm chambers (1 and 3) to a cold chamber (2) on one side with a smooth microchannel (1–2) and on the other side with a rough microchannel (3–2), a difference of pressure ($p_1 < p_3$) appears between the two chambers at the same temperature (Fig. 2.28).

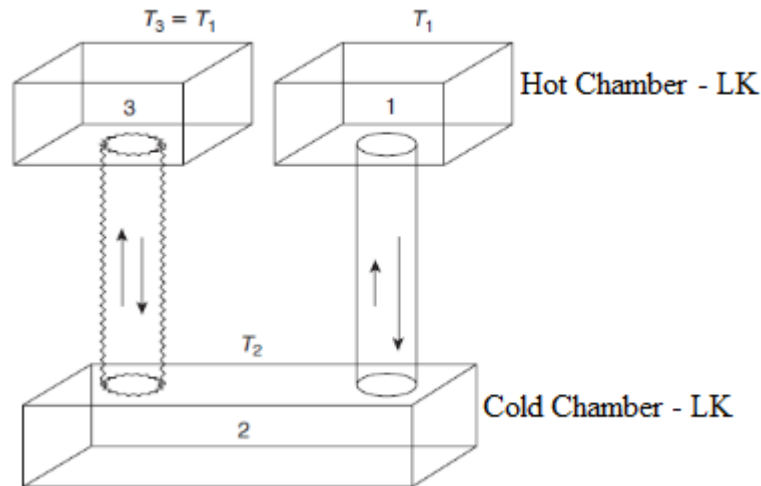


Fig. 2.28. One stage of an accommodation micropump.

Connecting in series several stages of that type, the pressure drops relative to each stage can be cumulated and **allow** the setup to reach **high vacuum levels**. The advantage of the accommodation pump is that it is operational without theoretical limitations concerning low pressures and contrary to the thermal transpiration pumping, the accommodation pumping does not require chambers with high dimensions. To compensate, the accommodation pump requires more stages to reach the same final pressure ratio. A pressure ratio $\Pi = 100$ requires 125 stages, whereas only 10 stages are enough for a transpiration pump with comparable temperatures. Lastly, no operational prototype was described in the literature so far, although the concept seems attractive. The highest-performance design was proposed by Hobson (1971; 1972); it requires a cold temperature $T_2 = 77$ K for an atmospheric temperature $T_1 = T_3 = 290$ K.

2.6. Heat transfer in microchannels

In the slip flow regime, heat transfer can be treated as well as mass transfer, providing the

energy equation (2.41)

(General explicit form of energy equation is added to the end of this chapter - LK)

$$\frac{\partial E}{\partial t} + \nabla \cdot \left(\frac{E + p}{\rho} \mathbf{J}_{EQ} - \sigma_{EQ} \cdot \mathbf{u} + \mathbf{q}_{EQ} \right) = 0 \quad (2.41)$$

and the

temperature boundary conditions (2.61)

$$T - T_{\text{wall}} = \frac{2 - \sigma_T}{\sigma_T} \frac{2\gamma}{\gamma + 1} \frac{k}{\mu c_v} \lambda \frac{\partial T}{\partial n} \Big|_w \quad (2.61)$$

are solved with the momentum equation and the slip-flow boundary conditions.

Such heat transfer problems have an analytical solution only in the simplest cases.

For example, if the compressibility of the flow can be neglected, the energy and the momentum equations are no longer coupled and,

if the flow is (hydrodynamically and thermally LK) developed,

longitudinal temperature gradients are constant and

longitudinal velocity gradients are zero (see example in Section 2.6.1.1).

In the general case,

analytical or semi-analytical solutions are not possible and

numerical simulations are required.

Depending on the accommodation coefficients, heat transfer can be found to increase, decrease or remain unchanged, compared to non-slip-flow conditions.

2.6.1. Heat transfer in a plane microchannel

2.6.1.1. Heat transfer for a fully developed incompressible flow

Let us consider the case of a plane microchannel (Fig. 2.10(a)), with a lower adiabatic wall ($q_w|_{y=-h} = 0$) and an upper wall with constant heat flux ($q_w|_{y=h} = q_h$).

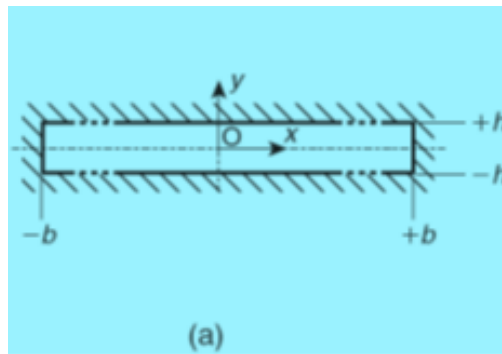


Fig. 2.10. Different microchannel sections.
(a) Plane channel limited by parallel plates ($b \gg h$)
(a part of original figure. LK)

The flow is assumed to be hydrodynamically and thermally developed and incompressible. Fluid properties are assumed uniform.

Noting $y^* = y/h$, the momentum equation reduces to

$$\frac{d^2 u_z}{dy^{*2}} = \frac{h^2}{\mu} \frac{dp}{dz} \quad (2.165)$$

and the energy equation to

$$\frac{k}{h^2} \frac{\partial^2 T}{\partial y^{*2}} = \rho u_z c_p \frac{\partial T}{\partial z} \quad (2.166)$$

neglecting viscous dissipation.

We introduce

$$\theta^* = \frac{T - T_{\text{wall}}}{q_h h / k} = \frac{4}{\text{Nu}} \frac{T - T_{\text{wall}}}{\bar{T} - T_{\text{wall}}} \quad (2.167)$$

with $q_h = \alpha(\bar{T} - T_{\text{wall}})$

and $\text{Nu} = \alpha D_H / k = 4\alpha h / k,$

noting α as the convection heat transfer coefficient.

The flow being thermally developed,

$$\partial\theta^*/\partial z = 0, \partial\text{Nu}/\partial z = 0 \text{ and } \partial q_h / \partial z = 0,$$

which implies $\partial T / \partial z = \partial \bar{T} / \partial z = \partial T_{\text{wall}} / \partial z.$

Consequently, the energy equation can be written as

$$\frac{k}{h^2} \frac{\partial^2 T}{\partial y^{*2}} = \rho u_z c_p \frac{\partial T}{\partial z} \quad (2.166)$$

$$\frac{q_h}{h} \frac{d^2 \theta^*}{dy^{*2}} = \rho u_z c_p \frac{d\bar{T}}{dz} \quad (2.168)$$

Moreover, the conservation of energy on a control volume of length dz and height $2h$ yields $2h\rho c_p \bar{u}_z d\bar{T} = -q_h dz$ and Eq. (2.168) reduces to

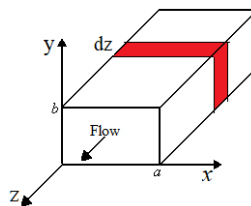
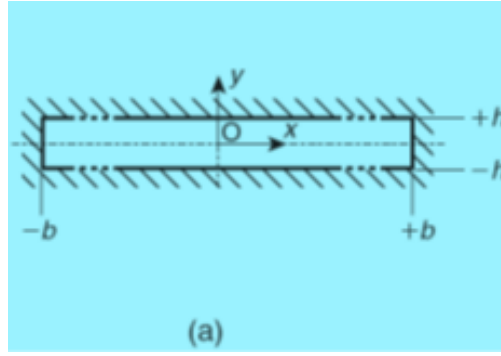


Figure is for “rectangular” microchannel, not for “plane” microchannel. However, they are similar. LK

$$\frac{d^2 \theta^*}{dy^{*2}} = -\frac{u_z}{2\bar{u}_z} \quad (2.169)$$

Neglecting thermal creep, the first-order boundary conditions are



$$\left. \frac{du_z}{dy^*} \right|_{y^*=0} = 0 \quad u_z|_{y^*=1} = -\xi^* \left. \frac{du_z}{dy^*} \right|_{y^*=1} \quad (2.170)$$

$$\left. \frac{d\theta^*}{dy^*} \right|_{y^*=-1} = 0 \quad \theta^*|_{y^*=1} = -\zeta^* \left. \frac{d\theta^*}{dy^*} \right|_{y^*=1} \quad (2.171)$$

with a dimensionless coefficient of slip

$$\xi^* = \xi/h = (\lambda/h)(2 - \sigma)/\sigma = 4Kn(2 - \sigma)/\sigma$$

and a dimensionless temperature jump distance

$$\zeta^* = \zeta/h = (\lambda/h)[(2 - \sigma_T)/\sigma_T][2\gamma/(\gamma + 1)]/Pr = 8[(2 - \sigma_T)/\sigma_T][\gamma/(\gamma + 1)][Kn/Pr],$$

where Pr is the Prandtl number.

The resolution of this set of equations gives the velocity distribution

$$u_z = -\frac{h^2}{2\mu} \frac{dp}{dz} (1 - y^{*2} + 2\xi^*) \quad (2.172)$$

and, with

$$\overline{u_z} = \frac{1}{2} \int_{-1}^1 u_z dy^*$$

$$\frac{u_z}{\overline{u_z}} = \frac{3(-y^{*2} + 1 + 2\xi^*)}{2(1 + 3\xi^*)} \quad (2.173)$$

$(\overline{u_z})$ is found before. LK)

The integration of Eq. (2.169) with the boundary conditions (2.171) yields

$$\theta^* = \varsigma^* + \frac{1}{1 + 3\xi^*} \left[\frac{1}{16} y^{*4} - \frac{3}{8} y^{*2} (1 + 2\xi^*) - \frac{1}{2} y^* (1 + 3\xi^*) + \left(\frac{13}{16} + \frac{9}{4} \xi^* \right) \right] \quad (2.174)$$

T and $\overline{T} = (1/2) \int_{-1}^1 (u_z/\overline{u_z}) T dy^*$ are calculated and

the **Nusselt number** is deduced as

$$\text{Nu}^{-1} = \frac{\varsigma^*}{4} + \frac{26 + 147\xi^* + 210\xi^{*2}}{140(1 + 3\xi^*)^2} \quad (2.175)$$

(General equations for Mean Fluid Velocity and Mean Fluid Temperature: LK)

Mean Fluid Velocity

$$u_m = \frac{\int_{A_c} \rho u dA_c}{\rho A_c}$$

Mean Fluid Temperature

$$T_m = \frac{\int_{A_c} \rho u c_p T dA_c}{\dot{m} c_p}$$

Equation (2.175) shows that **heat transfer** depends both on

the dimensionless coefficient of slip ξ^*

and on

the dimensionless temperature jump distance ς^* ,

that is, it depends on

the two accommodation coefficients σ and σ_T

as well as on the

Knudsen number Kn .

For classical values of the accommodation coefficients, the effect of the temperature jump is greater than the effect of velocity slip and the heat transfer decreases when rarefaction increases. The Nusselt number reduces to the classic value $70/13$ (Rohsenow, 1998) when $Kn = 0$.

Zhu et al. (2000) **extended** this solution to the problem of **asymmetrically heated walls** ($0 \neq q_0 \neq q_h$) by **combining** the **solutions** of **two sub-problems** similar to the above problem.

Inman (Karniadakis and Beskok, 2002) first studied the solution for a constant **symmetric heat flux** at the two walls.

The problem with **constant wall temperature** was treated by Hadjiconstantinou and Simek (2002), who extended the study to the transition regime, using DSMC. They showed that the slip flow prediction (with first-order boundary conditions) is in good agreement with the DSMC results for $Kn \leq 0.2$ and remains a good approximation beyond its expected range of applicability.

2.6.1.2. Heat transfer for a developing compressible flow

This problem was studied by Kavehpour et al. (1997), using the same first-order slip flow and temperature jump boundary conditions as in Section 2.6.1.1. The flow was developing both hydrodynamically and thermally and two cases were considered: uniform wall temperature or uniform wall heat flux. Viscous dissipation was neglected, but compressibility was taken into account. The numerical methodology was based on the control volume finite difference scheme. It was found that the Nusselt number was substantially reduced for slip flow compared with continuum flow.

2.6.2. Heat transfer in a circular microtube

Heat transfer in circular microtubes was simulated by Ameel et al. (1997), who considered a hydrodynamically fully developed incompressible flow, thermally developing with constant heat flux at the wall.

Viscous dissipation and thermal creep were neglected and first-order velocity slip and temperature jump conditions were used, with accommodation coefficients taken equal to unity.

Thermal creep, or transpiration phenomenon shows that a flow can be caused in the sole presence of a tangential temperature gradient, without any pressure gradient.

They found the thermally fully developed solution

$$\text{Nu}^{-1} = \frac{24\chi^2 - 16\chi + 3}{48(2\chi - 1)^2} \left(1 + \frac{24\gamma(\chi - 1)(2\chi - 1)^2}{(24\chi^2 - 16\chi + 3)(\gamma + 1)\text{Pr}} \right) \quad (2.176)$$

with $\chi = 1 + 4\text{Kn}$.

Equation (2.176) reduces to the no-slip value $\text{Nu} = 48/11$ when $\text{Kn} = 0$.
($\text{Nu} = 48/11 = 4.36 \text{ LK}$)

A semi-analytical solution is also given in Ameel et al. (1997) for the entrance region. For a Knudsen number $\text{Kn} = 0.04$, it is found that the Nusselt number for the fully developed flow is reduced 14% over that obtained with a no-slip boundary condition, with $\gamma = 1.4$ and $\text{Pr} = 0.7$.

Larrode et al. (2000) treated the case of constant temperature at the wall.

2.6.3. Heat transfer in a rectangular microchannel

Yu and Ameen (2001) studied the heat transfer problem for a thermally developing flow in a rectangular microchannel. The flow was assumed to be incompressible and hydrodynamically fully developed, and a constant temperature was imposed at the walls.

The velocity distribution was developed in series as discussed in Section 2.3.4.1 and the energy equation was solved by a modified generalized integral transform method.

A competition between ξ and ζ was pointed out, with a transition value β_c of $\beta = \xi/\zeta$ that depends on the aspect ratio a^* (Table 2.7).

Table 2.7

Transition values of $\beta = \xi/\zeta$ as a function of the aspect ratio, (from Yu and Ameen (2001)).

a^{*-1}	1	2	3	4	5	6	8	10	∞
β_c	0.67	0.50	0.38	0.32	0.29	0.28	0.27	0.25	0.20

For $\beta < \beta_c$, heat transfer is enhanced in comparison with the no-slip flow and if $\beta > \beta_c$, heat transfer is reduced.

Moreover, for $\beta < \beta_c$, heat transfer increases with increasing ξ^* and it decreases if $\beta > \beta_c$. (this is wrong, the correct one is $\beta > \beta_c$ LK)

According to the values available in the literature, it should be noted that β may vary in a wide range from unity to 100 or more, for actual wall surface conditions.

However, the value $\beta = 1.67$, which corresponds to a

diffuse reflection ($\sigma = 1$) and a total thermal accommodation ($\sigma_T = 1$)

for air ($\gamma = 1.4$; $Pr = 0.7$) is a representative value for

many engineering applications.

In this typical case, rarefaction effects result in a decrease of heat transfer, which can be as much as 40%.

2.7. Future research needs

Theoretical knowledge is currently well advanced for gas flows in microchannels.

However, there is yet a need for accurate experimental data, both for steady or unsteady gas microflows, with or without heat transfer. For example, in order to definitely validate the choice of the best boundary conditions in the slip flow regime, we need to isolate the influence of the accommodation coefficients, as it remains an open issue. Relationships between their values, the nature of the substrate and the micro-fabrication processes involved are currently not available. Theoretical investigations relative to unsteady or thermally driven microflows would also need to be supported by smart experiments.

2.8. Solved examples

Example 2.1

A microchannel with constant rectangular cross-section is submitted to a pressure-driven flow of argon. The inlet and the outlet pressures are $p_i=0.2$ MPa and $p_o = 25$ kPa respectively. The geometry is given in Fig. 2.10(c), with a width $2b = 20$ μm , a depth $2h = 1$ μm and a length $l = 5$ mm.

Assume a uniform temperature $T = 350$ K.

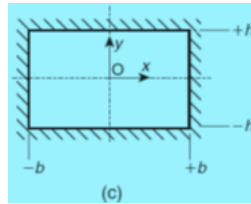


Fig. 2.10. Different microchannel sections.
(c) rectangular channel.

- (i) Calculate the Knudsen numbers Kn and Kn' at the inlet and at the outlet of the microchannel, assuming that the intermolecular collisions are accurately described by a variable soft spheres model.
- (ii) Compare with the values obtained from a hard spheres model.
- (iii) Assuming a diffuse reflection at the wall, calculate the mass flow rate through the microchannel.
- (iv) Calculate the mass flow rate increase (in %) due to slip at the wall.
- (v) What would be the underestimation (in %) of the mass flow rate given by a no-slip model, in the same conditions but with an inlet pressure $p_i=0.1$ MPa?

Solution:

(i) Calculate the Knudsen numbers Kn and Kn' at the inlet and at the outlet of the microchannel, assuming that the intermolecular collisions are accurately described by a **variable soft spheres model**.

(i) Inlet and outlet Knudsen numbers with aVSS model (Answer: $Kn'_i = 3.48 \times 10^{-2}$; $Kn_i = 1.83 \times 10^{-2}$; $Kn'_o = 2.78 \times 10^{-1}$; $Kn_o = 1.46 \times 10^{-1}$)

From Table 2.2, the temperature exponent of the coefficient of viscosity for argon (Ar) is $\omega = 0.81$, the exponent for the VSS model is $\alpha = 1.40$ and the dynamic viscosity under standard conditions is $\mu_0 = 211.7 \times 10^{-7}$ Pa s.

From Eq. (2.7), the viscosity at $T = 350$ K is:

$$\mu(T) = \mu_0(T/T_0)^\omega = 2.117 \times 10^{-5} (350/273.15)^{0.81} = 2.588 \times 10^{-5} \text{ Pa s.}$$

The molecular weight of argon is given in Table 2.2: $M = 39.948 \times 10^{-3}$ kg/mol and its gas constant is $R = R/M = 8.314511/39.948 \times 10^{-3} = 2.08 \times 10^2$ J/kg/K.

The mean free path is given by Table 2.1: $\lambda = k_2 \mu(T) \sqrt{RT/p}$, with

$$k_{2,VSS} = \frac{4\alpha(7 - 2\omega)(5 - 2\omega)}{5(\alpha + 1)(\alpha + 2)\sqrt{2\pi}} = 0.996$$

At the inlet,

$$\begin{aligned} \lambda_{i,VSS} &= k_2 \mu(T) \sqrt{RT/p_i} = 0.996 \times 2.588 \times 10^{-5} \times \sqrt{2.08 \times 10^2 \times 350/2 \times 10^5} \\ &= 34.8 \text{ nm} \end{aligned}$$

and at the outlet,

$$\begin{aligned} \lambda_{o,VSS} &= k_2 \mu(T) \sqrt{RT/p_o} = 0.996 \times 2.588 \times 10^{-5} \times \sqrt{2.08 \times 10^2 \times 350/2.5 \times 10^4} \\ &= 278 \text{ nm.} \end{aligned}$$

The Knudsen number based on the hydraulic diameter

$D_H = 4bh/(b + h) = 4 \times 10 \times 10^{-6} \times 0.5 \times 10^{-6}/((10 + 0.5) \times 10^{-6}) = 1.905 \mu\text{m}$ is $Kn = \lambda/D_H$, that is

$$\begin{aligned} Kn_i &= \lambda_i/D_H = 3.48 \times 10^{-8}/1.905 \times 10^{-6} = 1.83 \times 10^{-2} \text{ at the inlet and} \\ Kn_o &= \lambda_o/D_H = 2.78 \times 10^{-7}/1.905 \times 10^{-6} = 1.46 \times 10^{-1} \text{ at the outlet.} \end{aligned}$$

The Knudsen number based on the microchannel depth is $Kn' = \lambda/(2h)$, that is

$$\begin{aligned} Kn'_i &= \lambda_i/(2h) = 3.48 \times 10^{-8}/1 \times 10^{-6} = 3.48 \times 10^{-2} \text{ at the inlet and} \\ Kn'_o &= \lambda_o/(2h) = 2.78 \times 10^{-7}/1 \times 10^{-6} = 2.78 \times 10^{-1} \text{ at the outlet} \end{aligned}$$

(ii) Compare with the values obtained from a **hard spheres model**.

(ii) Inlet and outlet Knudsen numbers with a HS model (Answer: $\text{Kn}'_{\text{i,HS}} = 4.13 \times 10^{-2}$; $\text{Kn}_{\text{i,HS}} = 2.17 \times 10^{-2}$; $\text{Kn}'_{\text{o,HS}} = 3.30 \times 10^{-1}$; $\text{Kn}_{\text{o,HS}} = 1.73 \times 10^{-1}$)

With a HS model, $\omega = 0.5$ and the viscosity

$$\begin{aligned}\mu(T) &= \mu_0(T/T_0)^{0.5} = 2.117 \times 10^{-5} (350/273.15)^{0.5} \\ &= 2.396 \times 10^{-5} \text{ Pa s is underestimated of 7.4\%}.\end{aligned}$$

The coefficient $k_{2,\text{HS}} = 16/(5\sqrt{2\pi}) = 1.277$ (cf. Table 2.1).

Consequently,

$$\begin{aligned}\lambda_{\text{i,HS}} &= k_{2,\text{HS}} \mu(T) \sqrt{RT}/p_i = 1.277 \times 2.588 \times 10^{-5} \times \sqrt{2.08 \times 10^2 \times 350/2 \times 10^5} \\ &= 41.3 \text{ nm}\end{aligned}$$

and

$$\begin{aligned}\lambda_{\text{o,HS}} &= k_{2,\text{HS}} \mu(T) \sqrt{RT}/p_o = 1.277 \times 2.588 \times 10^{-5} \times \sqrt{2.08 \times 10^2 \times 350/2.5 \times 10^4} \\ &= 330 \text{ nm}\end{aligned}$$

which leads to:

$$\text{Kn}_{\text{i,HS}} = \lambda_{\text{i,HS}}/D_{\text{H}} = 4.13 \times 10^{-8}/1.905 \times 10^{-6} = 2.17 \times 10^{-2}$$

$$\text{Kn}_{\text{o,HS}} = \lambda_{\text{o,HS}}/D_{\text{H}} = 3.30 \times 10^{-7}/1.905 \times 10^{-6} = 1.73 \times 10^{-1}$$

$$\text{Kn}'_{\text{i,HS}} = \lambda_{\text{i,HS}}/(2h) = 4.13 \times 10^{-8}/1 \times 10^{-6} = 4.13 \times 10^{-2}$$

$$\text{Kn}'_{\text{o,HS}} = \lambda_{\text{o,HS}}/(2h) = 3.30 \times 10^{-7}/1 \times 10^{-6} = 3.30 \times 10^{-1}$$

The **overestimation** of the Knudsen number calculated from an **HS model**, compared with the more accurate solution obtained with a **VSS model**, is 28.2%, which corresponds to the ratio $(k_{2,\text{HS}} - k_{2,\text{VSS}})/k_{2,\text{VSS}} = 0.282$.

(iii) Assuming a diffuse reflection at the wall, calculate the mass flow rate through the microchannel.

(iii) Mass flow rate (Answer: $\dot{m} = 4.89 \times 10^{-12}$ kg/s)

The rectangular cross-section has a non-negligible aspect ratio $a^*=h/b=1/20=0.05>0.01$. Therefore, a rectangular model should be used. Due to the range covered by the Knudsen numbers, a second-order model (NS2) is required (cf. Fig. 2.7).

The mass flow rate is obtained from Eq. (2.157):

$$\dot{m}_{\text{NS2, rect}} = \frac{4 h^4 p_o^2}{a^* \mu R T l} \left[\frac{a_1}{2} (\Pi^2 - 1) + a_2 \text{Kn}'_o (\Pi - 1) + a_3 \text{Kn}'_o^2 \ln \Pi \right]$$

with $a_1 = 0.32283$, $a_2 = 1.9925$ and $a_3 = 4.2853$ given by Table 2.5 for a diffuse reflection ($\sigma = 1$). $\Pi = p_i/p_o = 200/25 = 8$ and

$$\begin{aligned} \dot{m}_{\text{NS2, rect}} &= 4(10^{-6})^4 (2.5 \times 10^{-4})^2 / (0.05 \times 2.588 \times 10^{-5} \times 2.08 \times 10^2 \times 350 \\ &\quad \times 5 \times 10^{-3}) \times [0.32283(8^2 - 1)/2 + 1.9925 \times 2.78 \times 10^{-1}(8 - 1) \\ &\quad + 4.2853 \times (2.78 \times 10^{-1})^2 \ln 8] = 4.89 \times 10^{-12} \text{ kg/s} \end{aligned}$$

(iv) Calculate the mass flow rate increase (in %) due to slip at the wall.

(iv) Mass flow rate increase (Answer: 44.9%)

From Eq. (2.158),

$$\begin{aligned} \dot{m}_{\text{NS2, rect}}^* &= \frac{\dot{m}_{\text{NS2, rect}}}{\dot{m}_{\text{ns, rect}}} = \left[1 + 2 \frac{a_2}{a_1} \text{Kn}'_o \frac{1}{\Pi + 1} + 2 \frac{a_3}{a_1} \text{Kn}'_o^2 \frac{\ln \Pi}{\Pi^2 - 1} \right] \\ &= 1 + 2 \frac{1.9925}{0.32283} 2.78 \times 10^{-1} \frac{1}{8 + 1} + 2 \frac{4.2853}{0.32283} (2.78 \times 10^{-1})^2 \frac{\ln 8}{8^2 - 1} = 1.449 \end{aligned}$$

which means that slip at the wall increases the flow rate of 44.9%.

(v) What would be the underestimation (in %) of the mass flow rate given by a no-slip model, in the same conditions but with an inlet pressure $p_i=0.1$ MPa?

(v) Underestimation of the mass flow rate (Answer: 46.7%)

If the inlet pressure is decreased to $p_i = 0.1$ MPa, the pressure ratio is $\Pi = 4$ and

$$\dot{m}_{\text{NS2, rect}}^* = 1 + 2 \frac{1.9925}{0.32283} 2.78 \times 10^{-1} \frac{1}{4 + 1} + 2 \frac{4.2853}{0.32283} (2.78 \times 10^{-1})^2 \frac{\ln 4}{4^2 - 1} = 1.877$$

Therefore, although the outlet Knudsen number is unchanged, the inlet pressure number is increased and rarefaction effects are enhanced. Slip at the wall now increases the flow rate of 87.7%. In others words, the underestimation of the mass flow rate with a no-slip model is 46.7%, since $(\dot{m}_{\text{NS2}} - \dot{m}_{\text{ns}})/\dot{m}_{\text{NS2}} = 1 - 1/\dot{m}_{\text{NS2}}^* = 1 - 1/1.877 = 0.467$.

Example 2.2

In a hard disk drive, the flow of gas between the hard disk and the read–write head can locally be comparable to a fully developed micro-Couette flow between parallel plane plates. The shield of the hard disk drive is sealed and the gas inside is pure nitrogen. The distance between the two plates is $2h = 0.2 \text{ } \mu\text{m}$, the read-write head is at $r = 3 \text{ cm}$ from the center of the disk, whose speed of rotation is $\dot{\omega} = 7200 \text{ rpm}$. One of the plates ($y = -h$) is fixed and the other ($y = h$) moves with a constant velocity U in the z -direction.

Assume a uniform temperature $T = 300 \text{ K}$, a uniform pressure $p = 90 \text{ kPa}$ and an accommodation coefficient $\sigma = 0.9$.

- (i) Write the velocity distribution as a function of the Knudsen number.
- (ii) Calculate the mass flow rate per unit width and compare it to the case of a no-slip Couette flow.
- (iii) Calculate the friction factor and compare its value to the one obtained with a no-slip assumption.

Solution:

(i) Write the velocity distribution as a function of the Knudsen number.

(i) Velocity distribution (Answer: $u_z = \frac{U}{2} \left(\frac{y^*}{4 \text{Kn}(2-\sigma)/\sigma + 1} + 1 \right)$)

Table 2.2 The equations of motion for an incompressible Newtonian fluid with constant viscosity in several coordinate systems	
Rectangular Coordinates (x,y,z)	
x-component	$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$

Or:

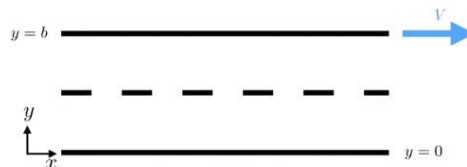
Fully Developed

Momentum x-component: $\rho \left[\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right] = \rho g_x - \frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$

Momentum y-component: $\rho \left[\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right] = \rho g_y - \frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right)$

Momentum z-component: $\rho \left[\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right] = \rho g_z - \frac{\partial P}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$

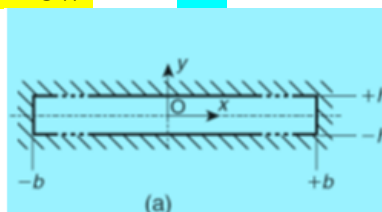
Conservation of Mass: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$



In fluid dynamics, **Couette flow** is the flow of a viscous fluid in the space between two surfaces, one of which is moving tangentially relative to the other. The configuration often takes the form of two parallel plates or the gap between two concentric cylinders. The flow is driven by virtue of viscous drag force acting on the fluid, but may additionally be motivated by an applied pressure gradient in the flow direction. The Couette configuration models certain practical problems, like flow in lightly loaded journal bearings, and is often employed in viscometry and to demonstrate approximations of reversibility.^[1] This type of flow is named in honor of Maurice Couette, a Professor of Physics at the French University of Angers in the late 19th century.

Couette flow: Low velocity flow

LK



The momentum equation reduces to $d^2 u_z / dy^2 = 0$,

and the boundary conditions are

$$u_z|_{y=-h} = ((2-\sigma)/\sigma)\lambda \, du_z/dy|_{y=-h} \quad \text{and} \\ u_z|_{y=+h} = U - ((2-\sigma)/\sigma)\lambda \, du_z/dy|_{y=+h}$$

Thus, the velocity profile is linear: $u_z = a_1 y + a_2$.

The boundary conditions yields

$$-a_1 h + a_2 = ((2-\sigma)/\sigma)\lambda a_1 \quad \text{and} \quad a_1 h + a_2 = U - ((2-\sigma)/\sigma)\lambda a_1,$$

which leads to

$$2 a_1 = U/(h + \lambda(2-\sigma)/\sigma) \quad \text{and} \quad 2 a_2 = U.$$

Then

$$u_z = U/2(y/(\lambda(2-\sigma)/\sigma + h) + 1).$$

With $y^* = y/h$ and $Kn = y/D_H = y/(4h)$, the velocity distribution is written as $u_z = U/2(y^*/(4Kn(2-\sigma)/\sigma + 1) + 1)$. The centerline velocity is independent of the slip at the walls and the velocity transverse gradient is reduced as rarefaction is increased.

(ii) Calculate the mass flow rate per unit width and compare it to the case of a no-slip Couette flow.

(ii) Mass flow rate

$$(\text{Answer: } \dot{m}' = (pM/RT)(\dot{\omega} r 2\pi/60)2h = 4.57 \times 10^{-6} \text{ kg/s/m})$$

The mass flow rate is

$$\dot{m} = \rho \overline{u_z} A = (p/RT) \overline{u_z} A \quad \overline{u_z} = 1/2 \int_{-1}^1 u_z \, dy^* = U/2.$$

Therefore, the mass flow rate per unit width is:

$$\dot{m}' = \frac{p}{RT} \frac{U}{2} 4h = \frac{pM}{RT} \frac{\dot{\omega} r 2\pi}{60} 2h \\ = \frac{9 \times 10^4 \times 28.013 \times 10^{-3} \times 7200 \times 3 \times 10^{-2} \times 2\pi \times 2 \times 10^{-7}}{8.3145 \times 300 \times 60} = 4.57 \times 10^{-6} \text{ kg/m}$$

The mass flow rate is the same as the one for a Couette flow without slip at the wall.

(iii) Calculate the friction factor and compare its value to the one obtained with a no-slip assumption.

(iii) Friction factor (Answer: $C_f^* = C_f/C_{f,ns} = 0.57$)

From Eq. (2.95),

$$C_f = 2 \overline{\tau_w} / (\rho \overline{u_z}^2) \quad \text{and} \quad \overline{\tau_w} = \tau_w = \mu \, du_z/dy$$

is uniform since the velocity profile is linear. With

$$\frac{du_z}{dy} = \frac{\dot{U}}{2h} \frac{1}{1 + 4 \text{Kn}(2 - \sigma)/\sigma}$$

we obtain:

$$C_f = \frac{4 \mu R T}{h p U (1 + 4 \text{Kn}(2 - \sigma)/\sigma)}$$

From Table 2.2, the temperature exponent of the coefficient of viscosity for nitrogen (N_2) is $\omega=0.74$, the exponent for the VSS model is $\alpha = 1.36$ and the dynamic viscosity under standard conditions is $\mu_0 = 1.656 \times 10^{-5}$ Pa s.

From Eq. (2.7), the viscosity at $T = 300$ K is:

$$\mu(T) = \mu_0 (T/T_0)^\omega = 1.656 \times 10^{-5} (300/273.15)^{0.74} = 1.775 \times 10^{-5} \text{ Pa s}$$

The molecular weight of nitrogen is given in Table 2.2: $M = 28.013 \times 10^{-3}$ kg/mol and its gas constant is $R = \sqrt{R/M} = 8.314511/28.013 \times 10^{-3} = 2.97 \times 10^2$ J/kg/K.

The mean free path is given by Table 2.1: $\lambda = k_2 \mu(T) \sqrt{RT}/p$, with

$$k_2 = [4 \alpha(7-2\omega)(5-2\omega)]/[5(\alpha+1)(\alpha+2)\sqrt{2\pi}] = 1.06.$$

Therefore,

$$\lambda = k_2 \mu(T) \sqrt{RT}/p = 1.06 \times 1.775 \times 10^{-5} \times \sqrt{2.97 \times 10^2 \times 300}/9 \times 10^4 = 62.6 \text{ nm}$$

and the Knudsen number based on the hydraulic diameter is:

$$\text{Kn} = \lambda/(4h) = 6.26 \times 10^{-8}/(4 \times 10^{-7}) = 1.56 \times 10^{-1}$$

The influence of rarefaction on the friction is then given by:

$$\begin{aligned} C_f^* &= C_f C_{f,ns} = C_f / C_{f(Kn=0)} = \frac{1}{1 + 4 \text{Kn}(2 - \sigma)/\sigma} \\ &= \frac{1}{1 + 4 \times 0.156(2 - 0.9)/0.9} = 0.57 \end{aligned}$$

The friction factor is reduced of 43%, due to slip at the walls.

2.9. Problems

Problem 2.1

A flow of helium is generated in a circular microtube by axial pressure and temperature gradients. The microtube has a length $l = 5$ mm and a uniform cross-section with a diameter $2r_2 = 10$ μm . In a section where the flow may be considered as locally fully developed, the gradients are $dp/dz = -5$ Pa/ μm and $dT/dz = 0.5$ K/ μm , for a pressure $p = 100$ kPa and a temperature $T = 400$ K.

Assume a tangential momentum accommodation coefficient $\sigma = 0.9$.

- (i) Write the velocity distribution as a function of the Knudsen number, in a dimensionless form: $u_z^*(r^*, \text{Kn})$.
- (ii) Calculate the part of the velocity increase due to slip at the wall.
- (iii) Calculate the part of the velocity increase due to thermal creep.

Problem 2.2

A flow of air in a rectangular microchannel is generated by a pressure gradient. The inlet pressure is $p_i = 3.5$ bar and the outlet pressure is $p_o = 150$ mbar. The temperature $T = 350$ K is uniform and the tangential momentum accommodation coefficient is $\sigma = 0.8$. The microchannel length is $l = 10$ mm and its uniform section has a depth $2h = 1.2$ μm and a width $2b = 4.8$ μm .

- (i) Calculate the mass flow rate with a plane flow assumption and a no-slip model.
- (ii) Calculate the mass flow rate with a plane flow assumption and a Navier–Stokes model with first-order slip boundary conditions.
- (iii) Calculate the mass flow rate with a plane flow assumption and a Navier–Stokes model with Deissler second-order slip boundary conditions.
- (iv) Calculate the mass flow rate with a plane flow assumption and a quasi-hydrodynamic model with first-order slip boundary conditions.
- (v) Calculate the mass flow rate with a no-slip model in a rectangular section.
- (vi) Calculate the mass flow rate with a Navier–Stokes model in a rectangular section with Deissler second-order slip boundary conditions.
- (vii) Calculate the deviations of the above different models, compared to the most accurate among them.

Problem 2.3

Consider a microchannel with a uniform cross-section whose width is very large compared with its depth $2h = 5$ μm . A pressure-driven flow of helium is generated, the lower wall ($y = -h$) is adiabatic and the upper wall ($y = h$) has a constant heat flux $q_h = -200$ W/ cm^2 . Consider a section far from the inlet, such that the flow in this section is thermally and hydrodynamically developed. The flow conditions in this section are such that the Knudsen number based on the hydraulic diameter is $\text{Kn} = 2 \times 10^{-2}$ and the accommodation coefficients are $\sigma = 0.9$ and $\sigma = 0.6$. The wall temperature $T_{\text{wall}} = 400$ K is measured with a micro-sensor.

- (i) Calculate the mean temperature $\bar{T}$ of the gas in the section.
- (ii) Calculate the temperature T_0 at the center of the section ($y = 0$).
- (iii) Compare with the case of a similar non rarefied flow.

Problem 2.4

Consider a microchannel, with a length $l = 5$ mm and whose uniform rectangular cross-section has a low aspect ratio: its width is $2b = 200$ μm and its depth is $2h = 1.5$ μm . A flow of nitrogen is generated with a pressure gradient. The inlet pressure is $p_i = 3 \times 10^5$ Pa and the outlet pressure is $p_o = 1 \times 10^5$ Pa. The temperature $T = 300$ K is uniform and the tangential momentum accommodation coefficient is $\sigma = 0.95$.

- (i) Calculate the mean velocity in a cross-section of the microchannel, as a function of the local pressure gradient and Knudsen number.
- (ii) Calculate the mass flow rate.
- (iii) Calculate the pressure value at the middle of the microchannel (at $z = l/2$) and compare it to the no-slip value.
- (iv) Calculate the maximal velocity reached in the microchannel outside the entrance region.
- (v) Calculate the maximal Mach number and justify the locally fully developed assumption.

Problem 2.5

A multistage Knudsen compressor is designed to locally tune the pressure level of air inside a microsystem. The geometry of this Knudsen compressor is given in Fig. 2.27. Ten chambers are connected via 9 series of 100 short microchannels. These microchannels have rectangular cross-sections with a depth $2h = 0.5$ μm and a width $2b = 20$ μm . Each chamber is assumed large enough for the flow inside to be considered as a continuum flow. The temperature in these chambers is $T_2 = 300$ K and each element is able to rise the temperature at the outlet of the microchannels to the value $T_1 = 600$ K. Chamber 10 is connected to a region, whose pressure $p_{10} = 10$ mbar is maintained constant by means of a primary pump. The initial temperature $T_{t=0} = T_2 = 300$ K is uniform before the elements heating. The initial pressure in each chamber is also uniform and is equal to the pressure in chamber 10: $p_{t=0} = p_{10} = 10$ mbar. For $t > 0$, heating of the elements can be regulated in order to tune the pressure in chamber 1 in the range 1 to 10 mbar.

- (i) Calculate the possible range covered by the Knudsen number in the microchannels during the pumping process and check that the free molecular regime assumption in the microchannel is valid.
- (ii) Calculate the maximal value of the mass flow rate between two adjacent chambers at the beginning of the pumping progress.
- (iii) Give the different reasons why the real initial mass flow rate is lower than the value calculated in (ii).
- (iv) Calculate the minimal pressure value that can be reached in chamber 1 at the end of the pumping process.

Problem 2.6

Consider a pressure-driven flow of different gases (He, O₂ or SO₂), through a microtube with rough walls, such that the accommodation coefficient can reasonably be assumed equal to unity. The microtube length is $l = 3$ mm and its inner diameter is $2r_2 = 1.2$ μm . The inlet pressure is $p_i = 2$ bar and the outlet pressure is $p_o = 1$ bar.

Assume that the intermolecular collisions are accurately described by a VSS model.

- (i) Calculate the mass flow rate through this microtube for the three different gases using a QHD1 model.
- (ii) Calculate the mass flow rate through this microtube for the different gases using a NS2 model with Deissler boundary conditions.
- (iii) Calculate the mass flow rate through this microtube for the different gases using a NS2 model with Chapman and Cooling boundary conditions.
- (iv) Calculate the mass flow rate through this microtube for the different gases using a NS2 model with Mitsuya boundary conditions.
- (v) Compare and comment on these different results.
- (vi) Propose an experimental protocol in order to check the more appropriate among the above models.

Appendix

Table 2.1 The continuity equation in several coordinate systems

General	Compressible	$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$
	Incompressible	$\nabla \cdot \mathbf{V} = 0$
Rectangular coordinates (x,y,z)	Compressible	$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) = 0$
	Incompressible	$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$
Cylindrical coordinates (r,θ,z)	Compressible	$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho w) = 0$
	Incompressible	$\frac{\partial v_r}{\partial r} + \frac{v_r}{r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial w}{\partial z} = 0$
Spherical coordinates (r,θ,φ)	Compressible	$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (\rho r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\rho v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (\rho v_\phi) = 0$
	Incompressible	$\frac{1}{r} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) + \frac{1}{\sin \theta} \frac{\partial v_\phi}{\partial \phi} = 0$

Table 2.2
The equations of motion for an incompressible Newtonian fluid
with constant viscosity in several coordinate systems

Rectangular Coordinates (x,y,z)	
x-component	$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$
y-component	$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right)$
z-component	$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$
Cylindrical Coordinates (r,θ,z)	
r-component	$\begin{aligned} \frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + w \frac{\partial v_r}{\partial z} &= f_r - \frac{1}{\rho} \frac{\partial p}{\partial r} \\ &+ \nu \left\{ \frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right] + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2} \right\} \end{aligned}$
θ-component	$\begin{aligned} \frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r} + w \frac{\partial v_\theta}{\partial z} &= f_\theta - \frac{1}{\rho r} \frac{\partial p}{\partial \theta} \\ &+ \nu \left\{ \frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right] + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} + \frac{\partial^2 v_\theta}{\partial z^2} \right\} \end{aligned}$
z-component	$\frac{\partial w}{\partial t} + v_r \frac{\partial w}{\partial r} + \frac{v_\theta}{r} \frac{\partial w}{\partial \theta} + w \frac{\partial w}{\partial z} = f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial w}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 w}{\partial \theta^2} + \frac{\partial^2 w}{\partial z^2} \right\}$
Spherical Coordinates (r,θ,φ)	
r-component*	$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\theta^2 + v_\phi^2}{r} \right) \\ = \rho f_r - \frac{\partial p}{\partial r} + \mu \left(\nabla^2 v_r - \frac{2v_r}{r^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} - \frac{2}{r^2} v_\theta \cot \theta - \frac{2}{r^2 \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right) \end{aligned}$
θ-component*	$\begin{aligned} \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} + \frac{v_r v_\theta}{r} - \frac{v_\phi^2 \cot \theta}{r} \right) \\ = \rho f_\theta - \frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left(\nabla^2 v_\theta + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r^2 \sin^2 \theta} - \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial v_\phi}{\partial \phi} \right) \end{aligned}$
φ-component*	$\begin{aligned} \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi v_r}{r} + \frac{v_\theta v_\phi}{r} \cot \theta \right) \\ = \rho f_\phi - \frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} + \mu \left(\nabla^2 v_\phi - \frac{v_\phi}{r^2 \sin^2 \theta} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} + \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial v_\theta}{\partial \phi} \right) \end{aligned}$
*In these relations: $\nabla^2 \equiv \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$	

Table 2.3 The energy equation for Newtonian fluids in several coordinate systems

Rectangular Coordinates	
$\boxed{\rho \frac{Di}{Dt} = \nabla \cdot (k \nabla T) + \frac{Dp}{Dt} + \dot{q} + \Phi}$	
$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$	
$\nabla \cdot (k \nabla T) = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)$	
$\Phi = 2\mu \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 \right. \\ \left. + \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 - \frac{1}{3} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)^2 \right]$	
Rectangular Coordinates (Perfect Gas)	
$\boxed{\rho c_p \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + \frac{Dp}{Dt} + \dot{q} + \Phi}$	
$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$	
$\nabla \cdot (k \nabla T) = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)$	
$\Phi = 2\mu \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 \right. \\ \left. + \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 - \frac{1}{3} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)^2 \right]$	
Rectangular Coordinates (Incompressible Fluid)	
$\boxed{\rho c \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + \dot{q} + \Phi}$	
$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$	
$\nabla \cdot (k \nabla T) = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)$	
$\Phi = 2\mu \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 + \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 \right]$	
Cylindrical Coordinates (Incompressible Fluid)	
$\boxed{\rho c \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + \dot{q} + \Phi}$	
$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + v_r \frac{\partial}{\partial r} + \frac{v_\theta}{r} \frac{\partial}{\partial \theta} + w \frac{\partial}{\partial z}$	
$\nabla \cdot (k \nabla T) = \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)$	
$\Phi = 2\mu \left\{ \left(\frac{\partial v_r}{\partial r} \right)^2 + \left[\frac{1}{r} \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right) \right]^2 + \left(\frac{\partial w}{\partial z} \right)^2 + \frac{1}{2} \left(\frac{\partial v_\theta}{\partial z} + \frac{1}{r} \frac{\partial w}{\partial \theta} \right)^2 \right. \\ \left. + \frac{1}{2} \left(\frac{\partial w}{\partial r} + \frac{\partial v_r}{\partial z} \right)^2 + \frac{1}{2} \left[\frac{1}{r} \frac{\partial v_r}{\partial \theta} + r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right]^2 \right\}$	

Table 2.3 Continued

Spherical Coordinates (Incompressible Fluid)

$$\boxed{\rho c \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + \dot{q} + \Phi}$$

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + v_r \frac{\partial}{\partial r} + \frac{v_\theta}{r} \frac{\partial}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial}{\partial \phi}$$

$$\nabla \cdot (k \nabla T) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 k \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(k \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right)$$

$$\Phi = 2\mu \left\{ \left(\frac{\partial v_r}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right)^2 + \frac{1}{r^2} \left(\frac{1}{\sin \theta} \frac{\partial v_\phi}{\partial \phi} + v_r + v_\theta \cot \theta \right)^2 + \frac{1}{2} \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 \right. \\ \left. + \frac{1}{2} \left[\frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} + r \frac{\partial}{\partial r} \left(\frac{v_\phi}{r} \right) \right]^2 + \frac{1}{2} \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right]^2 \right\}$$